

N73

Dato:

$$R = 0,3 \text{ m}$$

$$m = 1 \text{ kg}$$

$$v = 30 \text{ km/h}$$

Permette:

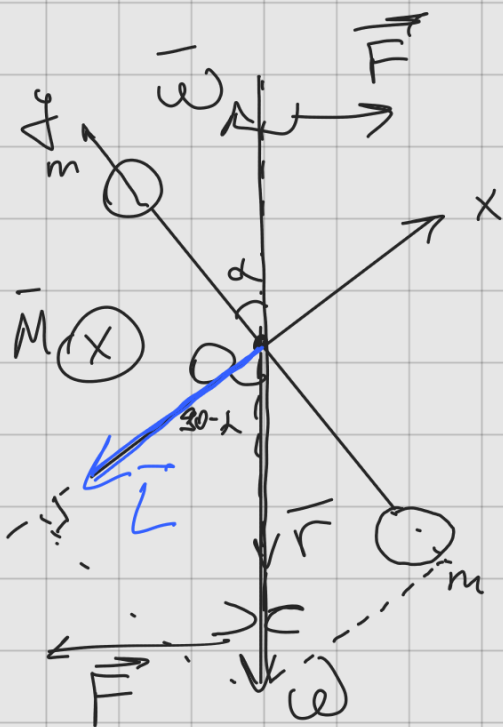
$$1) L = I\omega, I = \frac{2}{3} mR^2$$

$$2) \omega = \frac{v}{R}$$

$$3) L = \frac{2}{3} mR^2 \cdot \frac{v}{R} = \frac{2 mRv}{3} = \frac{1,6 \text{ kg} \cdot \text{m}^2}{\text{s}}$$

$$4) M = \Omega L = \frac{\Delta \alpha}{\Delta t} \cdot 1,6 = \frac{1}{0,1} \cdot 1,6 = 16 \text{ N} \cdot \text{m}$$

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$$\omega_x = -\omega \sin \alpha$$

$$\omega_y = -\omega \cos \alpha$$

$$I_x = \frac{2me^2}{4} = \frac{me^2}{2}$$

$$I_y = 0$$

$$|L_x| = \left| -\frac{me^2}{2} \omega \sin \alpha \right| = \left| -\frac{me^2}{2} \omega \sin \alpha \right| = |L|$$

$$M = \Omega L \sin(90-\alpha) = \underbrace{\Omega}_{\omega} L \cos \alpha = \omega L \cos \alpha$$

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Дано:

$$N = 7500 \text{ Дж/мин}$$

$$m = 12 \text{ Т}$$

$$\rho = 0,25 \text{ м}$$

$$R = 600 \text{ м}$$

$$v = 72 \text{ км/ч}$$

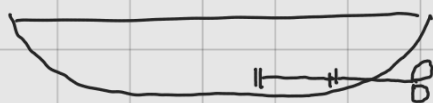
$$a = 1 \text{ м}$$

Найти: F

Решение:

$$1) \omega = 2\pi \nu = \frac{2\pi N}{60} = \frac{\pi N}{30} =$$

$$= 25\pi$$



$$2) L = I\omega = m\rho^2\omega$$

$$3) \Omega = \frac{v}{R} \Rightarrow M = \Omega L = \frac{m\rho^2\omega v}{R} \Rightarrow$$

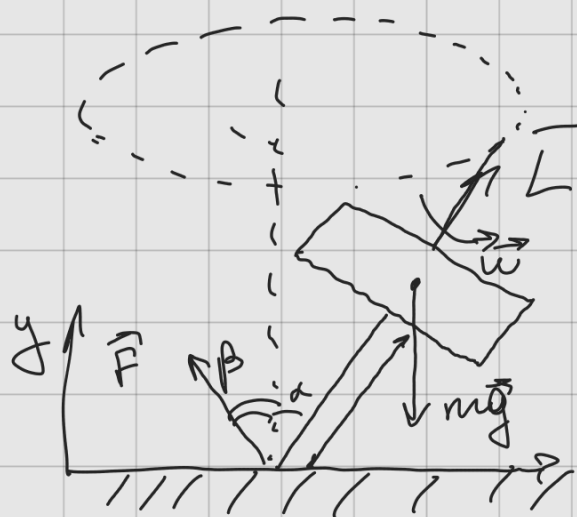
$$\Rightarrow F = \frac{m\rho^2\omega v}{Ra}$$

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Дано:

 m, α, a

Найти:

 β 

$$L = I\omega$$

$$M = mg \sin \alpha a$$

$$mg \sin \alpha a = \Omega \cdot L \cdot \sin \alpha$$

$$\omega - \text{известно} \Rightarrow$$

$$\Rightarrow \Omega = \frac{mga}{I \sin \alpha}$$

$$2) \text{ IЗ. H. } \kappa\alpha : F \sin \beta = m a_y ; F \cos \beta = mg$$

$$F \sin \beta = m \cdot \Omega^2 a \sin \alpha$$

$$\left. \begin{array}{l} \sin \beta = \frac{m \Omega^2 a \sin \alpha}{F} \\ \cos \beta = \frac{mg}{F} \end{array} \right\} \tan \beta = \frac{\Omega^2 a \sin \alpha}{g} =$$

$$= \frac{a \sin \alpha}{g}, \quad \frac{m^2 g^2 a^2}{I^2 \omega^2} = \frac{m^2 a^3 g \sin \alpha}{I^2 \omega^2}$$